

Incident Report: Simulated SSH Reconnaissance & Brute-Force Attack

Prepared by: Adebisi Damilola Abiola

Date: September 8, 2026

Assignment: Bincom Academy Cybersecurity Class - Assignment 1: Network Attack Detection & Reporting

1. Executive Summary

A controlled lab environment was built to simulate an external attacker performing reconnaissance and unauthorized access attempts against a network service. A Cowrie honeypot was deployed on an isolated Ubuntu VM to emulate a vulnerable SSH server. From a separate Kali Linux attacker VM, a full port scan was performed and multiple SSH login attempts were made against the honeypot while network traffic was captured for analysis. The exercise successfully demonstrated the full attack lifecycle, reconnaissance, exploitation attempt, and traffic capture, and produced evidence suitable for incident review.

2. Attack Summary

The attacker (Kali VM) first performed a full TCP port scan against the target to identify open services. This revealed two SSH-related services: the target's genuine SSH daemon on port 22, and a second SSH-like service on port 2222 later confirmed to be the Cowrie honeypot, impersonating a Debian host under the hostname 'svr04'. The attacker then initiated multiple SSH connection attempts against port 2222 using varied, non-legitimate credentials (e.g. root/[test password], admin/[test password]), consistent with a brute-force/credential-guessing pattern. Each session was accepted by the honeypot, which logged the attacker's commands (whoami, ls, cat /etc/passwd) before the session was terminated. All traffic, the initial scan, the TCP handshakes, and the SSH session negotiations was captured live in Wireshark for a full second scan and interaction window.

3. Technical Details

Attacker IP (Kali)	172.20.10.8
Target/Honeypot IP (Ubuntu, Cowrie)	172.20.10.10
Real SSH service	Port 22 - OpenSSH 8.9p1, Ubuntu
Honeypot SSH service	Port 2222 - Cowrie, spoofing OpenSSH 9.2p1 / Debian ('svr04')
Tools used	Nmap 7.98 (scanning), Wireshark 4.6.4 (capture), Cowrie (honeypot)
Scan type	Full TCP port scan with service/version detection (nmap -sV -p-)
Login attempts	Multiple, varied credentials (root, admin) against port 2222
Packets captured	355 packets, 0% dropped

4. Evidence

- Screenshot of Cowrie honeypot running and listening on port 2222
- Nmap scan results showing open ports 22 and 2222 with service versions
- Wireshark packet capture file: cowrie_attack_capture.pcapng (355 packets)
- Cowrie session logs recording attacker commands and credentials attempted

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Screenshot 2 - Nmap scan results showing open ports 22 and 2222

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kali@kali:~$
Session Actions Edit View Help

Ign1 http://http.kali.org/kali kali-rolling InRelease
Ign2 https://artifacts.elastic.co/packages/8.x/apt/stable InRelease
Ign3 http://http.kali.org/kali kali-rolling InRelease
Ign4 https://artifacts.elastic.co/packages/8.x/apt/stable InRelease
Ign5 http://http.kali.org/kali kali-rolling InRelease
Ign6 https://artifacts.elastic.co/packages/8.x/apt/stable InRelease
Ign7 http://http.kali.org/kali kali-rolling InRelease
Ign8 https://artifacts.elastic.co/packages/8.x/apt/stable InRelease
Temporary failure resolving 'artifacts.elastic.co'
Err1 http://http.kali.org/kali kali-rolling InRelease
Temporary failure resolving 'http.kali.org'
Err2 https://artifacts.elastic.co/packages/8.x/apt/stable InRelease
Temporary failure resolving 'artifacts.elastic.co'
Err3 http://http.kali.org/kali kali-rolling InRelease
Temporary failure resolving 'http.kali.org'
The package can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Failed to fetch http://http.kali.org/kali/dists/kali-rolling/InRelease Temporary failure resolving 'http.kali.org'
Warning: Failed to fetch https://artifacts.elastic.co/packages/8.x/apt/dists/stable/InRelease Temporary failure resolving 'artifacts.elastic.co'
Warning: Some index files failed to download. They have been ignored, or old ones used instead.

kali@kali:~$
kali@kali:~$ ping 172.20.10.10
PING 172.20.10.10 (172.20.10.10) 56(84) bytes of data.
64 bytes from 172.20.10.10: icmp_seq=1 ttl=64 time=0.4 ms
64 bytes from 172.20.10.10: icmp_seq=2 ttl=64 time=0.53 ms
64 bytes from 172.20.10.10: icmp_seq=3 ttl=64 time=0.58 ms
64 bytes from 172.20.10.10: icmp_seq=4 ttl=64 time=0.58 ms
64 bytes from 172.20.10.10: icmp_seq=5 ttl=64 time=0.38 ms
64 bytes from 172.20.10.10: icmp_seq=6 ttl=64 time=0.38 ms
64 bytes from 172.20.10.10: icmp_seq=7 ttl=64 time=0.38 ms
64 bytes from 172.20.10.10: icmp_seq=8 ttl=64 time=0.32 ms

--- 172.20.10.10 ping statistics ---
 8 packets transmitted, 8 received, 0% packet loss, time 5000ms
rtt min/avg/max/mdev = 0.2879/0.5204/14.631/3.964 ms

kali@kali:~$
kali@kali:~$ ping -v -c 1 172.20.10.10
Starting Nmap 7.90 ( https://nmap.org ) at 2026-09-07 12:34 -0100
Nmap scan report for 172.20.10.10
Host is up (0.038s latency).
Not shown: 65533 closed tcp ports (reset)
OS: STATE SERVICE VERSION
22/tcp open ssh OpenSSH 8.9p1 Ubuntu Ubuntu:15 (Ubuntu Linux; protocol 2.0)
22/tcp open ssh OpenSSH 8.9p1 Debian Debian:12 (Debian GNU/Linux; protocol 2.0)
MAC Address: 08:00:27:D6:E5:7D (Oracle VM VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:Linux;Linux_Kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.253 seconds

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Screenshot 3 - Wireshark packet capture of attack traffic

The screenshot displays the Wireshark network protocol analyzer interface. At the top, the status bar indicates the file is 'tcpdump.pcapng' and the capture is running. The main packet list pane shows a single packet (No. 1) of type TCP, source 172.20.10.8, destination 172.20.10.8, and port 443. The packet details pane shows the TCP header and options, including the SYN flag and the sequence number 172.20.10.8. The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol header, and TCP segment.

Screenshot 4 - Cowrie session logs recording attacker commands and credentials

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